Parmbir Singh Nandha

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BSc Computer Science (AI), Year 2 | 12-month placement from summer 2027 (software, AI, cyber)

CAREER PROFILE

Second-year Computer Science (AI) student at Brunel University London who builds software in Python, Java and TypeScript, including an AI finance tutor (retrieval-augmented generation, Next.js, FastAPI) and a phishing email analyser. Information Security intern on the HS2 London Tunnels project: supported a passwordless rollout for 1,000 staff and built a supplier risk-scoring workflow in Power Automate.

SKILLS

Languages: Python, Java, TypeScript; familiar with C++, SQL, JavaScript and Swift

Software and AI: Next.js, React, FastAPI, Flask, LangChain, LlamaIndex, ChromaDB, Ollama (local LLMs and RAG), NumPy, machine learning (K-Means, logistic regression, Q-Learning)

Security and IT: Microsoft Entra ID, Conditional Access, Microsoft Defender, Power Automate, Azure DevOps, risk assessment, threat modelling, networks and operating systems (university modules)

Engineering practice: Git, Maven, SSH, JUnit, PyTest, OO design, requirements specification, Agile/Scrum

Professional skills: communication (staff guides, queries and stakeholder meetings; weekly PAL sessions), teamwork (group projects of 10 and 4), initiative (took over unfinished requirements), budgeting (Society Treasurer)

WORK EXPERIENCE

Information Security Intern — SCS Railways (HS2 London Tunnels), Euston, London *June – Aug 2026*

- Supported the rollout of passwordless authentication to 1,000 staff: wrote the passkey setup guide, sent rollout communications and answered staff queries.
- Configured Named and Trusted Locations for Conditional Access policies in Microsoft Entra ID in a live environment, and reviewed contractor access so people only had what they needed.
- Investigated suspicious sign-ins (such as unexpected overseas logins) using Entra ID and Defender, and contributed to security risk assessments and stakeholder meetings.
- Built a Power Automate workflow that scored suppliers' risk assessments against security certifications (ISO 27001, Cyber Essentials) and sent an automatic reply with their risk status; raised and tracked findings as Azure DevOps tickets.
- Took part in third-party threat-modelling workshops ("how could an attacker get in?") and documented and presented the case for stopping personal VPN use on the company network.

AI/ML Virtual Internship Program — ArithMatrix Technologies (remote) *Sep – Oct 2026*

- **Customer segmentation:** segmented 4,325 customers from 540k+ transactions using RFM and K-Means (k=4); found that 15.9% of customers generate 63.9% of revenue (£5.4M). *Python*
- **Spam classifier:** TF-IDF and logistic regression model with 98.84% accuracy and 93.89% recall (123 of 131 spam caught, 4 false positives in 903 legitimate messages). *Python*
- **Q-Learning agent:** built a 10x10 grid-world in NumPy and trained a tabular Q-Learning agent from scratch; win rate rose from 0.4% to 99.2%, with the path verified optimal using BFS. *Python, NumPy*

PROJECTS

NetValor: AI Money Coach *Oct 2026 – Present*

- Building a full-stack personal finance learning app with an AI co-pilot that answers from cited lesson content using retrieval-augmented generation. The back end runs locally: lesson search, streaming cited answers and a Python calculation engine, so the model never does the maths. Front end in development from Figma designs. *Next.js, TypeScript, Python, FastAPI, ChromaDB, Ollama*

HookLine: Phishing Email Analyser *Sep 2026 – Present*

- Phishing analyser that parses .eml files, runs 13 red-flag checks (SPF/DKIM/DMARC, Reply-To mismatches, look-alike domains) and rate-limited, cached VirusTotal lookups, then returns a Safe, Suspicious or Malicious verdict with reasons. Caught 48 of 50 real phishing emails; I analysed each miss. Some checks were tuned on the same 50, so this is a best case. *Python, Flask, VirusTotal API*

The Archive: Local RAG Chatbot *Apr – May 2026*

- Fully local chatbot that answers from a user's own documents using semantic retrieval and similarity-score gating, falls back to Wikipedia and streams cited answers. *Python, FastAPI, LlamaIndex, ChromaDB, Ollama*

UK Tick Tracker: Backend API *Oct – Nov 2026*

- Backend built for a placement technical task: imports tick-sighting data with timeouts and retries, cleans and de-duplicates it, and serves search by date range and location plus sightings per region and over time. 56 automated tests, README and video walkthrough. *Python, FastAPI, SQLite, PyTest*

Spybot: Secure Message Delivery Robot *Jan – Mar 2026*

- Wrote a 35-requirement SRS, OO design, JUnit tests and a hardware test plan for a SwiftBot app using QR authentication and Morse code; deployed over SSH to a Raspberry Pi. Graded A*. *Java, Maven, Raspberry Pi*

Hillingdon x Brunel Hackathon 2026 *Apr 2026*

- Led UI/UX design in a team of 4 for a council reporting system for road hazards and fly-tipping; presented to judges from Hillingdon Council, Microsoft, Brunel and ICS.AI. Recognised by Microsoft. *Python, Figma*

CertCheck: Staff Certificate Tracker (CS2701 Group Project) *Oct 2026 – Present*

- Team of 4 designing a web app that helps small businesses track staff certificate expiry (UN SDG 8). Wrote the user personas, empathy map, Lean Canvas and project plan.
- Planning a 3-week front-end sprint using Agile/Scrum, including the product backlog and sprint backlog. *Agile, Scrum*

Group Project: Simon Says for SwiftBot *Oct – Dec 2025*

- Team of 10 building a Java memory game, working in code sprints at least twice a week with Waterfall and Git. Completed the requirements specification when a teammate was unavailable. *Java, SwiftBot API, Notion*

POSITIONS OF RESPONSIBILITY

PAL Leader (Peer Assisted Learning) — Brunel University London *Sep 2026 – Present*

- Appointed to run weekly peer-led study sessions for first-year Computer Science students, breaking down programming and core module concepts for groups.

Widening Access Student Ambassador — Brunel University London *Appointed Oct 2026*

- Appointed as a student ambassador for Brunel's widening access work.

Co-founder and Treasurer — Brunel Cybersecurity Society *June 2026 – Present*

- Manage the society's finances: set and track the annual budget, administer Students' Union funding and sponsor funds, and allocate budgets for workshops, socials and a student hackathon.
- Designed the society's live website (brunelcs.org) in Figma for desktop and mobile (homepage, committee, events and CTF pages), defined its visual style, worked with the committee to deliver a responsive site and reviewed the build against the designs.

EDUCATION

Brunel University of London — BSc (Hons) Computer Science (AI) *2025 – 2029*

- Year 1: 2:1 (provisional), including Software Implementation (A*), Software Design (A+) and Introductory Programming (A).
- Year 2 (in progress): Software Development and Management, Algorithms, Networks and Operating Systems, Usability Engineering, Level 5 Group Project.

West London College — BTEC L3 Extended Diploma in IT: Distinction, Distinction, Distinction *2023 – 2025*

- Units included Programming, Software Testing, Data Modelling, Cyber Security and Incident Management and Mobile Apps Development.

CERTIFICATIONS

- **Anthropic:** Claude Code in Action (Jul 2026) and Claude Code 101 (Aug 2026): AI-assisted development and CI/CD automation.
- **Codecademy:** Intro to Java and Learn C++ (Sep 2026).
- **LinkedIn Learning:** Introduction to NLP and LLMs (Sep 2026); Python Data Modeling and Model Evaluation Metrics (Jul 2026).
- **Google:** Build Real World AI Applications with Gemini and Imagen skill badge (Apr 2026). **IBM:** Artificial Intelligence Fundamentals (Nov 2025).
- **LinkedIn Learning:** continuous delivery (CI/CD) course, in progress.

INTERESTS

Interests and community: football and running (Brunel Running Society), gaming and Brunel's Punjab Society. Volunteered at a community kirtan event (Nov 2025), serving free food (langar) to the public.